



Suites 4-7, 193 Minna Jane Drive
Charlottetown, PE C1E 0C4

 info@redsandspm.com

 www.redsandspm.com

 +1 782-377-7813

Radiofrequency Ablation (RFA) for Chronic Pain: Patient Information Leaflet

What is RFA?

Radiofrequency Ablation (RFA) is a minimally invasive procedure that uses heat from radio waves to interrupt pain signals from specific nerves to your spinal cord. This is performed by inserting special needles close to the target nerves and heating them. It is recommended for some types of chronic pain that has not improved with conservative treatments like oral medications, cortisone injections or physical therapy.

The aim is to reduce pain, improve function, delay surgery and reduce the need for pain medications.

How does RFA work?

Mechanism

A needle is guided near the target nerve using imaging (e.g., X-ray or ultrasound). Radio waves heat the nerve tissues, creating a small lesion that blocks pain signals.

Types of RFA

Conventional:

Continuous heat (80°C) – used to create a lesion to provide pain relief from 6 to 12 months.

Pulsed:

Intermittent energy (lower heat, ~40°C) – used to modulate nerve function to reduce some types of Nerve (neuropathic) pain.

Duration of effect and common Indications for RFA

One session of RFA can provide 6-12 months of pain relief. The procedure needs to be repeated when pain returns as the nerves repair themselves.

The following are some common indications for RFA. There could be other uses of RFA as well.

Lumbar Facet Joint pain:

Usually pain which is worse with bending backwards and rotating your back.

Cervical Facet Joint pain:

Pain in the neck, shoulders, upper back and sometimes arms, most commonly caused after whiplash injuries.

Occipital Headaches:

Pain originates from a nerve in the neck called Third Occipital Nerve and can cause back of the head pain sometimes radiating to the back of your eyeball.

Sacroiliac Joint pain:

Pain in the very low back, behind the hips sometimes this pain radiates to the hip joints and groin.

Coccygeal Pain:

This is commonly known as tail bone pain. It is worse on sitting and gets better when you stand and move.

Joint Pain:

RFA can be used to reduce knee or hip pain.

Neuropathic Pain:

Sometimes, when the nerves have been partially damaged, they misfire and cause burning or shooting pain. At times touch or other non-painful stimuli feel as pain. Pulsed RFA can help ease such pain.

How will I know if I am candidate for RFA?

Eligibility:

Your pain specialist will discuss with you if your pain can be treated with RFA.

- You must have temporary pain relief from diagnostic nerve blocks as detailed below.
- RFA is not suitable for pregnant patients and those with infections or bleeding disorders.
- Please inform your doctor if you have a pacemaker, defibrillator or a spinal cord simulator implanted in your body.

If you are a possible candidate, then there is a two-step process.

Step 1 – Diagnostic Blocks:

Numbing injections are performed first to confirm the nerve source and to check if these injections reduce the pain by at least 80%. Sometimes two sets of diagnostics blocks are performed on separate occasions, before deciding to proceed with the RFA. If you get 80% pain relief with diagnostic blocks there are 70-80% success rate that you will achieve similar results after the RFA.

If you have good results with diagnostic blocks, you would prepare as follows and the second step will be scheduled on a separate occasion.

Preparation:

- Stop blood thinners – your doctor will advise you depending on the type of blood thinner.
- If you are having the procedure under sedation, please do not eat solid food for 6 hours prior to your scheduled procedure time. You can drink clear fluids including black coffee or tea without milk up to two hours prior to the procedure.
- If you are diabetic please do not take your morning pills or short acting insulin dose.
- Arrange transportation (you **CANNOT** drive home).

Step 2 – RFA procedure:

Duration:

30–90 minutes, depending on sites treated.

Steps:

- If you have chosen to have the procedure under sedation you will get an IV line established and mild sedation will be given in a monitored environment to keep you comfortable and safe.
- Local anesthesia numbs the skin.
- Needle placement guided by fluoroscopy or ultrasound.
- Electrical stimulation test to confirm that the needle is in a safe position and there is no twitching in unexpected muscles. Your doctor will explain to you as the procedure goes along.
- After confirmation of safe needle position under imaging and testing as above heat is applied for 90 to 120 seconds per nerve after injecting further local anesthetic to complete the procedure.

Risks and Side Effects

In addition to the risks listed below there is a 20-30% chances of failure to get pain relief either due to technique failure or individual variation in the location of the nerves in human bodies.

Common:

- Temporary soreness, bruising, or numbness at the needle site (resolves in days), icing can help.
- Neuritic (burning or shooting) pain usually not lasting more than 2-4 weeks. Especially common after RFA procedures in the neck. You might have to take medicines like Gabapentin or Pregabalin to numb this pain till it resolves.
- RFA in the neck can cause Ataxia or Vertigo (balance issues) for a couple of weeks, the chances of this happening are very high.

Rare:

- Infection or bleeding (<2%).
- Nerve injury or worsened pain.
- Rarely, RFA can cause muscle weakness in neck, back or leg, depending upon where the RFA is performed.

After the Procedure

Immediate Care:

- Rest for 24 hours; avoid driving or strenuous activity, especially if you had sedation for the procedure.
- Remove the bandage applied to the needle instruction sites at 24 hours.
- Ice the injection site for swelling.

Recovery:

- Pain relief may start in 1–4 weeks, sometimes longer with the neck procedures.
- Physical therapy is recommended to strengthen muscles as pain reduces.

Long-Term:

- Nerves can regenerate in 6–18 months; RFA can be repeated if pain returns.
- Greater than 80% of patients reduce pain medication use.

Frequently Asked Questions

Q: Is RFA permanent?

A: No, nerves often regenerate. Repeating treatments can maintain relief.

Q: Can RFA treat other conditions?

A: Yes! It is also used for knee and hip arthritis, trigeminal neuralgia, and some types of cancer pain.

Q: What if I have an implant (e.g., pacemaker)?

A: Inform your doctor, some implants may require precautions.

When to Call Your Doctor?

Seek immediate help for:

- Severe redness/swelling at the needle site.
- Chills and fever
- New weakness/numbness in limbs, neck or back.

Questions? Contact Your Care Team

Before/after procedure:

 Pain Clinic: 782-377-7813

 Info@redsandspm.com

In an emergency

→ Call 911, or visit your local emergency department